

De-kit Bundles Back to Bins

One episode takes **four kits** apart and puts every part back where it came from, using the two-arm tabletop robot. Each kit is one kit card and four SKU parts, held by one band, with one kit label stuck on top. The table begins with a box holding the four kits on the right, four empty SKU bins and an empty card spot on the left, a clear work area in the center, and a cutter, a waste bin, and a tally card with four empty squares. It ends with the box empty, every part back in its own bin, the four cards back in one stack, the four cut bands and four kit labels in the waste bin, and four chips on the tally card.

The workflow is: bring one kit to the center of the table, cut its band, take the parts apart, peel the kit label off the top part, put every part back in the bin it came from, put the card back on the card stack, and put one chip in the next empty tally square. That workflow runs once per kit, one kit at a time, until all four kits are taken apart and tallied.

Nothing is ever pulled apart before the band is cut, and no part goes into a bin while the kit label is still on it.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the full tabletop, the four SKU bins and the card spot on the left, the center work area, the cutter spot and the waste bin, the box of kits, and the tally card.
3. The center work area is seen from above, so every cut, every peel, and every part can be seen.
4. Both arms are at home with grippers open.
5. The table is clear except for the bins, the card spot, the cutter, the waste bin, the box, the tally card, the chip cup, and robot hardware.

Materials checklist

1. Place the box, holding four finished kits, in the back-right zone. Each kit is one kit card with four SKU parts on it, one band around the middle, and one kit label stuck on the top part.
2. Each kit holds one part from each of the four SKUs. No kit holds two parts of the same SKU.
3. Every part carries its own printed SKU. On the top part of each kit that SKU is hidden under the kit label and shows again once the label is off.
4. Stage four empty SKU bins in the back-left supply zone. Each bin is marked with one SKU, and the four marks match the four SKUs in a kit.
5. Keep the front-left card spot empty. The kit cards are stacked here as they come out of the kits.
6. Lay one cutter flat on the cutter spot at the front right, handle toward the right arm. It has an open hook blade, so it cuts a band that is lifted into the hook and cannot cut a part lying flat.
7. Place one open waste bin, empty, at the front-right corner. It takes the cut bands and the peeled kit labels only.
8. Place the tally card beside the box. It has four squares, numbered 1 to 4, and all four squares begin empty.
9. Place four count chips, one per kit, in a small cup next to the tally card.
10. Keep the center work area clear and dry.
11. Before collection, confirm that the hook of the cutter goes under one band, that one kit label peels off a part in one piece, and that one chip sits flat inside one tally square.

Workspace layout

- **Back-left supply zone:** the four empty SKU bins, one SKU mark per bin
- **Front-left card zone:** the card spot, empty at the start, where the kit cards are stacked
- **Center work area:** cutting, taking apart, and peeling
- **Front-right supply zone:** the cutter on its spot and the open waste bin
- **Back-right output zone:** the box of kits, emptied left to right, plus the tally card and the chip cup

Arm assignments

- **Left gripper:** brings the kit to the center, holds the kit still while the band is cut, holds the part still while the kit label is peeled, and holds the stack square while parts are taken off it.
- **Right gripper:** takes the cutter and cuts the band, throws the band and the label away, lifts the parts off one at a time, carries every part to its bin, puts the card back, and places the count chip.

The assignments are fixed for the whole episode: the left gripper holds, and the right gripper cuts, carries, sorts, and tallies. Neither gripper does the other one's work.

Vocabulary

- **SKU:** one kind of part. Each bin is marked with one SKU only.
- **Part:** one single item that came out of a kit. It carries its own printed SKU.
- **Kit:** one kit card, four parts, one band, and one kit label, held together as one block.
- **Kit card:** the printed card that lists the four SKUs. It sits under the parts and is not a part.
- **Band:** the rubber band around the middle of the kit that holds it together.
- **Cut band:** a band cut through in one place, so it comes off the kit in one piece.
- **Kit label:** the sticker on the top part that shows the kit number. It hides the SKU printed on that part.
- **Part SKU:** the SKU printed on a part. On the top part it can only be read once the kit label is off.
- **Loose parts:** the four parts lying apart on the work area, none on top of another and none touching the card.
- **Origin bin:** the bin whose mark is the same as the SKU printed on the part in the gripper. Every part has one origin bin and no other.
- **Sorted:** a part resting inside its origin bin, with nothing left on it.
- **Cutter:** the tool with the open hook blade that cuts one band. It is held by its handle only.
- **Cutter spot:** the place the cutter lies flat on before and after every cut. The cutter is never left anywhere else.
- **Waste bin:** the open bin that takes cut bands and peeled kit labels. Nothing else goes in it.
- **Card stack:** the kit cards on the card spot, laid one on top of another, printed side up.
- **Next kit:** the first kit in the box counting from left to right. Kits leave the box in that order.
- **Tally card:** the card with four numbered squares. One square is filled for each kit taken apart.
- **Count chip:** one small chip placed in a tally square to show one kit is done.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower in:** carrying level over the bin, lowering straight down until the thing is resting, and releasing once it is resting. Nothing is dropped in from above the rim.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or rolling.
- **Release point:** over the center work area, over the origin bin, over the card spot, over the waste bin, or over the tally square the thing in the gripper is going to. A gripper opens nowhere else.

Steps

Run Steps 1–6 once for every kit, one kit at a time. After the fourth kit is sorted and tallied, end the episode with Step 7.

Step 1: Bring one kit to the center

Goal: one whole kit lies flat in the center work area, kit label facing up, and the other kits stay in the box.

1. Find the **next kit** in the box, counting from left to right.
2. With the **left gripper**, lift that kit straight up out of its box space. Lift one kit only.
3. Carry it level to the center work area, clear of the box wall and the tabletop.
4. Lay it flat with the kit label facing up and release it.
5. Do not touch the kits still in the box.

Check: one whole kit lies flat in the center, label up and band still on, and the other kits sit as they were in the box. The center work area holds nothing else.

Step 2: Cut the band

Goal: the band is cut in one place and thrown away, and the kit still lies flat with all its parts in place.

2.1 Hold the kit and take the cutter

1. With the **left gripper**, hold the kit down flat on the work area, away from the band.
2. Press just hard enough to keep the kit still. Do not squeeze a part out of shape.
3. With the **right gripper**, take the **cutter** by its handle from the cutter spot and bring it to the kit.

Expected state: the kit lies flat under the left gripper, and the cutter is held by its handle in the right gripper.

2.2 Cut the band

1. Slide the hook of the cutter under the band on the top face of the kit.
2. Lift the band a little away from the kit, so only the band is inside the hook.
3. Close the cutter until the band cuts through.
4. Cut the band in one place only. Never cut with the blade resting on a part, on the card, or on the table.

Expected state: the band is cut through in one place, the kit still lies flat, and no part is cut, scratched, or moved out of the stack.

2.3 Throw the band away and park the cutter

1. Carry the cutter back to the **cutter spot**, lay it flat, and release it.

2. With the **right gripper**, take the cut band by one end and pull it clear of the kit in one slow motion.
3. Carry the band to the **waste bin**, lower it in, and release it.
4. The left gripper keeps holding the kit until the band is clear of it.

Check: the band is in the waste bin in one piece, the cutter lies flat on its spot, and the kit lies in the center as one flat stack with nothing holding it.

Step 3: Take the parts apart

Goal: the four parts lie apart in the center work area, with the kit card empty and lying flat beside them.

1. With the **left gripper**, hold the stack square at its middle so it does not slide.
2. With the **right gripper**, lift the top part straight up off the stack and lay it flat in the center work area, clear of the stack.
3. Lift one part at a time, working down the stack from the top.
4. Lay each part apart from the others, so no part rests on another part and no part rests on the card.
5. Keep the part that carried the kit label where you can tell it from the rest.
6. Stop when the kit card is bare. The card is not a part and is not lifted yet.

Check: four loose parts lie apart in the center, the kit card lies bare and flat, and nothing is still stacked or stuck together. If a part came up with another, lay both down and take them one at a time.

Step 4: Take the kit label off

Goal: the kit label is off the part and in the waste bin, and the SKU printed on that part can be read again.

1. With the **left gripper**, hold that part flat on the work area, label facing up.
2. With the **right gripper**, take one corner of the **kit label**.
3. Peel the label off in one slow motion, keeping it flat and pulling it back across itself.
4. Peel the whole label off. Do not leave a corner or a strip of it on the part.
5. Carry the label to the **waste bin**, lower it in, and release it.
6. Read the SKU that shows on the part. That SKU says which bin the part goes back to.

Check: no kit label is left on any part, the label is in the waste bin, and the SKU printed on that part can be read. If the label tears, peel the rest of it off before going on.

Step 5: Sort the parts back to their bins

Goal: all four parts rest in their origin bins, and the kit card is back on the card stack.

1. With the **right gripper**, lift one part from the center work area. Lift one part only.
2. Read the SKU printed on it and find the bin marked with that same SKU. That is its **origin bin**.
3. Carry the part level to that bin, lower it in through the open top, and release it once it is resting.
4. Do the same for the other three parts, one at a time, until the center work area holds only the kit card.
5. With the **right gripper**, lift the kit card and lay it on the **card spot**, printed side up, on top of the cards already there.
6. Take every part in and out from above, over the open top of the bin. Do not drag a part over a bin wall.

Check: every part sits in the bin marked with its own SKU, no bin holds a part from another SKU, the card stack is neat and printed side up, and the center work area is clear.

Step 6: Mark the tally card

Goal: one count chip sits in the next empty tally square, so the number of chips matches the number of kits taken apart.

1. With the **right gripper**, take one count chip from the cup. Take one chip only.
2. Find the next empty tally square, counting up from square 1.
3. Lay the chip flat inside that square and release it.
4. Count the chips on the tally card and count the empty spaces in the box. The two numbers must be the same.
5. Place the chip only after the parts are in their bins and the card is on the stack, never before.

Check: there is one chip for each kit taken apart, each chip lies flat inside its own square, and the squares are filled in order from square 1. If the two counts do not match, fix the tally before starting the next kit.

If a kit remains in the box, rerun Steps 1–6 on the next kit. Continue to Step 7 only after the fourth kit is sorted and tallied.

Step 7: End the episode

Goal: recording ends after all four kits have been cut open, taken apart, unlabeled, sorted, and tallied.

1. Confirm the box is empty and every space is bare.

2. Confirm each of the four bins holds four parts of its own SKU only, and no part is left on the table.
3. Confirm four kit cards lie in one stack on the card spot, printed side up.
4. Confirm the waste bin holds four cut bands and four kit labels, and that the cutter lies flat on its spot.
5. Confirm four chips sit in the tally card, one per square, and the chip cup is empty.
6. Return both arms home with grippers open. Homing is the last thing the arms do.
7. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. Take the parts out of the four bins and build four kits again: one card, four parts, one from each SKU, card at the bottom.
2. Put one fresh band around the middle of each kit and stick one fresh kit label on the top part, covering its printed SKU.
3. Put the four kits back in the box, one per space, filled from left to right, label up.
4. Empty the waste bin of the cut bands and the peeled labels.
5. Take the four chips out of the tally card and put them back in the cup.
6. Wipe any label glue off the parts, and wipe the center work area clean and dry.
7. Inspect the parts, cards, cutter, box, bins, tally card, and chips. Replace anything torn, bent, cracked, blunt, or worn out.
8. Confirm the four bins and the card spot are empty and all four tally squares are empty, then run both Setup checklists.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Violation: Wrong order

- **Visible cue:** a part is pulled off before the band is cut, a part goes into a bin while the kit label is still on it, the card goes back before the parts are sorted, or a chip is placed before the kit is fully sorted.
- **SOP rule broken:** Steps 1–6 (cut, take apart, unlabel, sort, then tally, in that order).
- **Coaching note:** cut it, split it, peel it, sort it, tally it. Finish the step you are in before you start the next one.

Violation: More than one kit worked at once

- **Visible cue:** a second kit is lifted out of the box, or parts of two kits lie in the center work area at the same time.
- **SOP rule broken:** Steps 1–6 (work one kit at a time, all the way through the tally card).
- **Coaching note:** finish one kit through Step 6, then take the next one out of the box.

Violation: Band torn, pulled, or rolled off instead of cut

- **Visible cue:** the band is stretched over the end of the kit, rolled off, snapped by pulling, or comes off whole with no cut in it.
- **SOP rule broken:** Step 2 (cut the band with the cutter in one place, then pull it clear).
- **Coaching note:** the band is always cut first. Pulling a band off drags the parts out of line.

Violation: Cut made on or near a part

- **Visible cue:** the blade rests on a part, on the kit card, or on the table when it closes, the band is not lifted into the hook first, or a part comes out cut, nicked, or scratched.
- **SOP rule broken:** Step 2 (lift the band away from the kit so only the band is inside the hook).
- **Coaching note:** hook it, lift it, then close. Nothing but the band goes inside the hook.

Violation: Cutter not returned to its spot

- **Visible cue:** the cutter is left in the work area, in a bin, in the box, on the card stack, or still in the gripper when the next step begins.
- **SOP rule broken:** Step 2.3 (the cutter goes back flat on the cutter spot before the band is pulled clear).

- **Coaching note:** the tool goes home as part of the step, not later.

Violation: Band or label not thrown away

- **Visible cue:** a cut band or a peeled kit label is left on the table, in a bin, in the box, on the card stack, or on the tally card.
- **SOP rule broken:** Steps 2.3 and 4 (the cut band and the peeled label go straight into the waste bin).
- **Coaching note:** waste goes to the waste bin, and nowhere else, before the next move.

Violation: Parts not fully taken apart

- **Visible cue:** two parts are still stacked, a part is still lying on the kit card, or more than one part is lifted off the stack in one grasp.
- **SOP rule broken:** Step 3 (lift one part at a time and lay each one apart from the others).
- **Coaching note:** one part per lift, laid down clear of the rest. If two come up, lay both down and take them again.

Violation: Kit label left on a part

- **Visible cue:** a part goes into a bin with the kit label still on it, or with a corner or a strip of the label still stuck to it.
- **SOP rule broken:** Step 4 (peel the whole kit label off before the part is sorted).
- **Coaching note:** the part is not ready for its bin until its own SKU can be read.

Violation: Part damaged while the label is peeled

- **Visible cue:** the printed SKU comes away with the label, the part is bent, or its surface is torn or lifted by the peel.
- **SOP rule broken:** Step 4 (hold the part flat and peel the label back across itself in one slow motion).
- **Coaching note:** peel it low and slow, back over itself, never straight up.

Violation: Part put in the wrong bin

- **Visible cue:** a part goes into a bin whose mark is not the SKU printed on that part, or two SKUs end up in one bin.
- **SOP rule broken:** Step 5 (every part goes back into its origin bin, and each bin holds one SKU only).
- **Coaching note:** read the SKU on the part, then read the mark on the bin, then let go.

Violation: More than one part carried at once

- **Visible cue:** two parts are lifted together toward the bins, or a part is dropped into a bin while another is still in the same gripper.

- **SOP rule broken:** Step 5 (lift and carry one part at a time).
- **Coaching note:** one part, one bin, one trip.

Violation: Kit card lost, thrown away, or left out

- **Visible cue:** the kit card goes into a SKU bin or the waste bin, is left in the center work area, or is still under a part when the tally chip is placed.
- **SOP rule broken:** Step 5 (the card goes back on the card spot, printed side up, once the parts are sorted).
- **Coaching note:** the card is not waste and is not a part. It goes on the card stack.

Violation: Part dropped or dragged

- **Visible cue:** a part or a kit falls, scrapes the tabletop, drags over a bin wall or the box wall, or is dropped in from above the bin rim.
- **SOP rule broken:** Steps 1, 3, and 5 (lift clear, carry level, and lower in until the thing is resting).
- **Coaching note:** lift clear, carry level, cross above a bin rather than through it.

Violation: Work done outside the center work area

- **Visible cue:** a band is cut, a kit is taken apart, or a label is peeled over the box, over a bin, over the card stack, or over the waste bin instead of in the center work area.
- **SOP rule broken:** Steps 2–4 (bring the kit to the center first and cut, take apart, and peel it there).
- **Coaching note:** kit to the center first, and keep every cut and every peel over it.

Violation: Tally does not match the box

- **Visible cue:** no chip is placed for a kit taken apart, two chips are placed for one kit, a chip is placed before the kit is sorted, a square is skipped, or a chip lands outside a square.
- **SOP rule broken:** Step 6 (place one chip per kit, flat inside the next empty square, in order from square 1, and only after the kit is sorted).
- **Coaching note:** sort the kit, then place one chip, then check that the two counts match.

Violation: Bins, box, card stack, or tally card displaced

- **Visible cue:** a gripper, an arm, or a carried part pushes, slides, or tips a SKU bin, the card stack, the cutter spot, the waste bin, the box, the chip cup, or the tally card, or knocks a placed chip out of its square.
- **SOP rule broken:** Steps 1–6 (cut, take apart, sort, and tally without disturbing the containers, the tally, or work completed earlier).
- **Coaching note:** approach everything from above on a clear path and keep the arm off the walls.

Violation: Wrong arm used

- **Visible cue:** an action given to one gripper is done by the other, including the left gripper taking the cutter, cutting the band, peeling the label, carrying a part to a bin, or placing the count chip.
- **SOP rule broken:** Steps 1–6 (the left gripper holds; the right gripper cuts, carries, sorts, and tallies).
- **Coaching note:** left gripper holds, right gripper works. The roles do not swap in this task.

Violation: Wrong episode ending

- **Visible cue:** recording stops before all four kits are sorted and tallied and the end state is confirmed, an arm is not home, a gripper is closed, or an arm performs another action after homing.
- **SOP rule broken:** Step 7 (sort and tally all four kits, confirm the completed state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by task execution are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- The cutter is blunt or will not close, so a band will not cut.
- A kit label will not peel off in one piece because the label itself is bad.
- A part, kit card, box, bin, tally card, or chip is defective or damaged before the episode starts.
- A kit reaches the table with a part missing, a SKU doubled, or no band on it. Reset error, not a violation.

Annotation subtasks (from SOP)

Subtasks 1–11 repeat once per kit, four times in the episode. Subtask 12 occurs once, at the end.

1. Bring one kit from the box to the center work area
2. Take the cutter from its spot
3. Cut the band on one kit
4. Return the cutter to its spot

5. Put the cut band in the waste bin
6. Lift one part off the stack and lay it apart
7. Peel the kit label off the top part
8. Put the kit label in the waste bin
9. Put one part in its origin bin
10. Put the kit card back on the card stack
11. Place one count chip in the next empty tally square
12. Confirm the end state, return both arms home, and end the episode